

EXPERIMENT NO. 8

AIM: To develop an interactive web application using HTML, CSS, and JavaScript, implementing DOM manipulation, JavaScript event handling, form validation, dynamic calculations, product filtering, cart management, checkout interaction, and dynamic user feedback.

TOOLS / SOFTWARE REQUIRED

- Visual Studio Code (VS Code)
- Google Chrome
- HTML5
- CSS3
- JavaScript (ES6)

THEORY

1. HTML Structure and Forms

HTML is used to create the basic structure of a web application. In this practical, HTML elements are used to create the user interface for the gym admission section and the StyleHub shopping interface. The gym interface contains fields for collecting user details and membership information, while the shopping interface contains navigation elements, search, filters, product cards, cart, checkout, and order confirmation sections.

Example:

```
<input type="text" id="name" placeholder="Enter your name">
```

The input element allows the user to enter information, which can later be accessed and processed using JavaScript.

2. CSS Styling and Responsive User Interface

CSS is used to control the appearance, layout, spacing, typography, colors, buttons, cards, forms, and modal windows of the application. In this practical, CSS is used to create a structured and user-friendly interface for the gym form and the StyleHub shopping page.

The shopping interface uses product cards, navigation menus, filter sections, pop-up dialogs, cart panels, and checkout forms. CSS helps maintain a consistent visual layout and improves the overall user experience.

3. DOM Manipulation

The Document Object Model (DOM) represents an HTML document as a tree-like structure of elements. JavaScript can access these elements and modify their content, properties, styles, classes, and values dynamically.

In this experiment, JavaScript uses DOM methods such as `getElementById()`, `querySelector()`, and `querySelectorAll()` to access and update elements.

Example:

```
const searchInput = document.querySelector("#search");  
const productContainer = document.querySelector(".product-container");
```

These statements select elements from the webpage so that JavaScript can dynamically update the displayed content.

DOM manipulation is used to update product listings, display product details, manage the shopping cart, show checkout information, calculate totals, and display successful registration or order messages.

4. Event Handling

An event is an action that occurs on a webpage, such as typing in a search box, selecting a filter, clicking a button, changing a form value, or submitting a form.

JavaScript provides event listeners to detect these actions and execute a function.

Common events used in this experiment are:

- input – Occurs when the user enters or changes text.
- change – Occurs when the value of a filter, checkbox, or selection changes.
- click – Occurs when the user clicks a button, product, cart control, or navigation element.
- submit – Occurs when a form is submitted.
- reset – Occurs when a form is reset.

Example:

```
searchInput.addEventListener("input", function() {  
    // update the displayed products  
});
```

The input event allows the product list to be updated dynamically according to the user's search text.

5. Product Search and Filtering

Product filtering is used to help users find products according to selected conditions. In the StyleHub interface, products can be viewed using categories and filters such as gender, category, and other available options.

JavaScript reads the selected filter values and compares them with the product data. Only products matching the selected conditions are displayed.

Similarly, the search box allows the user to enter a product name or keyword. The input event can be used to filter the product array and dynamically update the product cards without reloading the webpage.

This makes the shopping interface interactive and easier to navigate.

6. Product Details and Cart Management

The shopping application provides a product details section where the user can view information about a selected product before adding it to the cart.

When the user selects a product, JavaScript dynamically displays its details such as product name, price, discount, rating, and other available information. The user can then add the product to the shopping cart.

The cart stores the selected products and allows the user to review the items before checkout. JavaScript updates the cart contents and calculates the corresponding amount dynamically.

7. Price, Discount and Total Calculation

The shopping interface performs dynamic calculations for the selected products. When products are added to the cart, JavaScript calculates the subtotal, discount, delivery charge, and final total.

For example:

Subtotal = Sum of product prices

Total = Subtotal – Discount + Delivery Charge

The calculated values are displayed dynamically in the cart and checkout sections. In the observed interface, the order summary shows the selected item, subtotal, discount, delivery charge, and total amount.

8. Form Validation

Form validation is the process of checking whether the information entered by the user is complete and follows the required format before an operation is completed.

In this practical, validation is applied to the gym admission form and the checkout form. Important information such as name, contact details, selected options, delivery address, and required fields are checked before proceeding.

If required information is missing or invalid, the user is asked to correct the entered data. This prevents incomplete form submission and improves the reliability of the application.

9. Checkout, Order Placement and Dynamic Feedback

The checkout section allows the user to enter or confirm delivery information and review the order summary before placing the order.

When the user clicks the place-order button, JavaScript processes the selected cart items and displays an order-success message. The interface shown in the practical includes an order confirmation dialog with an option to view the receipt.

A receipt can display information such as Order ID, Date, selected product, quantity, subtotal, discount, delivery charge, total amount, and payment method. This provides clear feedback to the user after the order is completed.

10. Dynamic User Interaction

JavaScript makes the complete application interactive by responding to user actions without requiring a page reload.

The application provides dynamic features such as:

- Product search
- Product filtering
- Product details display
- Add-to-cart functionality
- Cart updates
- Price and total calculation
- Checkout interaction
- Form validation
- Order confirmation
- Receipt display
- Gym admission success message
- Reset or submit-again functionality

These features demonstrate how HTML, CSS, and JavaScript can be combined to create an interactive and user-friendly web application.

CONCLUSION

The experiment was successfully performed to develop an interactive web application using HTML, CSS, and JavaScript. DOM manipulation and JavaScript event handling were used to dynamically interact with webpage elements. Product search, filtering, product details, cart management, checkout, price calculation, form validation, order confirmation, and dynamic user feedback were implemented. The practical demonstrated how JavaScript can be used with HTML and CSS to create interactive, responsive, and user-friendly web applications.